

PFM 204 Course Outline

PFM 204 — Government Budget Analysis

Course Outline

School	02 — Macroeconomics and Public Finance
Level	200 (analyst / practitioner)
Hours	30 (indicative band 24–32)
Shape	5 modules · 10 sessions · 6 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101. No prior Julia is assumed; the laboratories introduce what is needed. Familiarity with a government budget classification is an advantage but is taught from first principles. PFM 203 may be taken before or after: the two courses meet at the FY0 baseline.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-24 · 12-month cycle

What this course is

PFM 204 teaches the AICTEM Budget module: a 8,337-line Julia implementation across 16 layers that imports an approved-budget workbook immutably, preserves the full ten-level official hierarchy, classifies every dollar analytically as well as administratively, maps every line to exactly one AICTEM sector through a ten-tier resolver, reconciles the result exactly to source, and publishes stable packages to the medium-term framework, to sixteen sector models, to a computable general equilibrium model and to a results framework.

It is built on the implemented module and its published run: a warehouse of 31 tables holding the approved Utopia FY2026/27 budget — 84.392 tn across 27,658 lines, 159 votes and 531 projects.

The boundary rule. BUDGET owns source fiscal truth and detailed classification.

Who it is for

Budget directorate and accounting officers' staff; fiscal data and statistics units; analysts in planning commissions and central banks who consume budget data; sector planning officers who receive a package rather than a spreadsheet; and EDIS developers maintaining the module or its interfaces.

What this course does not claim

This course does not forecast. Forward estimates, sector ceilings and scenarios belong to PFM 203.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	What the budget module is; The source and the hierarchy;	01–03	01–02	9.1
M2	Reconciliation Classification: appropriation against economics	04–05	03–04	8.4
M3	Ownership, sectors and the centre	06	02	1.7
M4	Execution, financing, projects and geography	07–08	02, 04	3.4
M5	The handovers	09–10	05–06	7.4
	Total	10	6	30.0

The 10 sessions and 6 laboratories, in teaching order, and the subject matter of each:

M1 What the budget module is; The source and the hierarchy; Reconciliation — Sessions 01–03 · 9.1 h · Laboratory 01–02

- **01 Source truth, and what this module is** — source truth, and why a module owns it; scope, non-scope and the boundary rule; the pipeline, and the gate in the middle; the nine rules the module will not break
- *Lab 01 Build the warehouse and read the verdict* — run the pipeline end to end, read the reconciliation record and the data-quality verdict, and establish for yourself that the warehouse contains exactly the money the workbook contains
- **02 The hierarchy, the components and the schema** — the source, held immutably; the ten-level official hierarchy; the seven financing components; the star schema, and more
- *Lab 02 Trace a figure to its cell* — take a published sector total, trace it through the bridge to the fact table and out to a workbook cell, and explain why a particular line is in the sector it is in
- **03 Reconciliation and the gate** — what reconciliation means here; blocking against non-blocking; the control totals; the data-quality scorecard

M2 Classification: appropriation against economics — Sessions 04–05 · 8.4 h · Laboratory 03–04

- **04 Appropriation is not economics** — appropriation is not economics; the analytical classes; capital forming and asset preservation

- *Lab 03 Classify a budget analytically* — reproduce the analytical classification, compute the capital-forming and asset-preservation totals, and quantify the error a model reading ‘development’ as ‘investment’ would make
- **05 Mapping at the lowest reliable level** — the ten-tier resolver; mapping governance and the review queue; primary ownership and secondary links; the sector picture, and the centre
- *Lab 04 Audit the mapping* — read the ten-tier resolver’s output, work the review queue in order of amount at stake, and write a manual override with a name and a reason

M3 Ownership, sectors and the centre — Session 06 · 1.7 h · Laboratory 02

- **06 Cash, execution and the chain** — the cash plan; the execution chain

M4 Execution, financing, projects and geography — Sessions 07–08 · 3.4 h · Laboratory 02, 04

- **07 External finance, counterpart and projects** — external financing; counterpart funding and implementation risk; the project registry
- **08 What the module does not know** — geography, and why a project has no coordinates; what the module does not know

M5 The handovers — Sessions 09–10 · 7.4 h · Laboratory 05–06

- **09 The handovers** — the sector packages; the MTEF package; the CDCGE shock: four buckets; the MEL linkage, and more
- *Lab 05 Build the CDCGE shock* — reproduce the four buckets, prove they sum to the budget, and quantify the two errors the classification prevents
- *Lab 06 Publish and consume a package* — publish the MTEF baseline and the sector packages, read a package as a consumer would, and demonstrate the dependency trap
- **10 Reading a budget honestly** — reading a budget honestly; ten common errors; A worked budget brief

What a graduate can do

1. Explain why one module must own source fiscal truth, and apply the boundary rule between Budget, MTEF and FINEX.
2. Read and reproduce a ten-level budget hierarchy and the grain of its fact table.
3. Name the seven financing components, their appropriation class and the trap each carries.
4. Reconcile a budget exactly at three levels, and classify a failed check as blocking, non-blocking or a defect.
5. Read a data-quality scorecard and act correctly on `safe_to_model`.
6. Distinguish an appropriation class from an analytical class, and quantify the error of confusing them.
7. Identify capital formation and asset preservation, and say which downstream number each becomes.
8. Read a ten-tier mapping resolver’s output, work a review queue by amount at stake, and write a governed override.
9. Apply primary ownership, and demonstrate the double count that secondary links prevent.
10. Read a cash plan for what it is, and distinguish the five stages of the execution chain.

The 15 formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	87 items across 10 modules
Laboratory exercises	30 %	6 laboratories, equally weighted, marked on method, interpretation, honesty about limits and communication
Applied assignments	25 %	reconciliation and verdict note; analytical classification note; mapping audit; handover note
Final assessment	30 %	Three hours, open book: definitions and structure; calculation; interpretation; reporting

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been recorded with its actual value: an unchecked result is no result.

Materials and set-up

A 151-page technical manual; 10 lecture decks and 6 laboratory decks; a laboratory manual, instructor guide and learner workbook; a 87-item quiz bank, 4 assignments and a final assessment with its marking guide; 7 datasets; 4 code exercises; 4 case studies; and the lms package. The README lists every file.

Every figure in this course is traceable to a published source or a committed model output; the sources are registered in `COURSE_VERSION.json` and the limitations in `COURSE_BUILD_REPORT.md`. Laboratories need only the extracts shipped in `08_Data/`.

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